

ABHIJEET GUPTA

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ML / AI Engineer | CS PhD Candidate | PyTorch | LLM Evaluation | RL | Computer Vision

SUMMARY

CS PhD candidate graduating May 2027 with research and industry experience in controlled model evaluation, reinforcement learning, computer vision, production data systems, and multi-GPU inference. Public repositories distinguish published claims, reconstructed implementations, and newly reproduced results.

TECHNICAL SKILLS

ML: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, Gymnasium, OpenCV, MediaPipe, CNNs, LSTMs, reinforcement learning, multilabel learning

Research engineering: Python, NumPy, pandas, SciPy, statsmodels, Slurm/HPC, multi-GPU inference, Docker, Kubernetes, GitHub Actions, deterministic experiments

EXPERIENCE

Graduate Research Assistant | University of Dayton | Aug 2023 - Present

- Builds controlled ML evaluation pipelines spanning adversarial reinforcement learning, computer vision, and reasoning-model analysis; coordinates reproducible Slurm and multi-GPU experiments.
- Co-authored peer-reviewed ML/CV research and mentors junior researchers on experiment design, PyTorch implementation, statistical validation, and publication artifacts.

Data & Systems Engineering Intern | Dayton Phoenix Group | Jan 2025 - Present

- Built Python data pipelines and KPI reporting for manufacturing systems; applied statistical process control and anomaly analysis to production metrics.

Software Engineer | Walmart Global Tech via Kitestring | Apr 2023 - Aug 2023

- Built Docker/Kubernetes CI/CD workflows for a Node.js point-of-sale application and Python validation tooling for migration of more than one million records to Azure Cosmos DB.

ML / Product Intern | Walmart | Jun 2022 - Aug 2022

- Developed a Random Forest supply-chain classifier and integrated heterogeneous operational data sources with leakage-aware validation.

SELECTED RESEARCH PROJECTS

Reasoning Model Failure Analysis | Private research repository

- Designed controlled evaluations across six reasoning models to separate reasoning-length effects from forced re-entry interventions; original artifacts remain private.
- Implemented multi-GPU inference and structured behavioral/mechanistic result capture. No private dataset or participant material is published.

Adversarial RL Feature Selection | Public reference reconstruction

- Built a deterministic perturbation benchmark comparing all features, mutual information, entropy, and KL-divergence selection with repeated-seed uncertainty.
- Local reproduction shows mutual information outperforming entropy and KL selection, while using all features remains strongest; published-paper results are explicitly separate.

Mouse Brain Cell Segmentation | Public reference reconstruction

- Implemented synthetic fluorescence generation, background correction, connected-component segmentation, Dice/IoU evaluation, object-count diagnostics, and qualitative artifacts.
- Deterministic benchmark achieved mean Dice 0.905 and IoU 0.827; public biological-dataset evaluation remains NOT_RUN pending paper and dataset review.

Virtual Yoga Instructor | Public reference reconstruction

- Implemented normalized joint-angle scoring, corrective feedback, landmark-noise robustness, and local geometry-engine latency evaluation.
- Measured 0.027 ms scoring latency excluding camera and landmark inference; end-to-end MediaPipe latency remains NOT_RUN and is not implied by this result.

Multi-Output Career Prediction | Public reference reconstruction

- Created a privacy-preserving synthetic multilabel benchmark with logistic and random-forest baselines, hamming loss, macro/micro F1, and subgroup reporting.
- Random forest reached 0.763 micro F1; results validate the reconstructed benchmark only.

GROVE Construction Safety Evaluation | Public research companion

- Packaged permitted GROVE evidence with component ablations, paired statistical tests, threshold sensitivity, failure attribution, and machine-readable claim traceability.

EDUCATION

PhD, Computer Science | University of Dayton | Expected May 2027

Research: LLM evaluation and interpretability, reinforcement learning, computer vision, and reproducible machine learning.

MS, Computer Science | University of Texas at Arlington | 2022

BE, Computer Science and Engineering | RTM Nagpur University | 2019